# Cloud Security

## (CST-109)

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# Assignment

| | Assignment 1: Load Balancer System | Assignment 2: Distributed File System |
|---|---|---|
| **Goal** | Efficiently distribute requests | Efficiently store & retrieve data/files |
| **Focus** | Request routing | Data storage & management |
| **Main Problem** | Which server should handle a request? | Where is the data stored and how to retrieve it? |
| **Client Interaction** | Sends many requests (up to 1M) | Uploads & downloads files |
| **Servers Do** | Process requests (stateless) | Store file chunks (stateful) |
| **Key Component** | Load Balancer | Metadata Server |
| **Data Handling** | No data storage | File chunking + storage |
| **Replication** | Not required (optional concept) | Mandatory (for fault tolerance) |
| **Failure Handling** | Redirect traffic to alive servers | Retrieve data from replicas |
| **Evaluation Metric** | Load distribution, response time | Data availability, reliability |
| **Cloud Platforms** | Private | Private |

# Assignment

| | Assignment 3: Secure Cloud Application Architecture |
|---|---|
| **Task** | **Objective** |
| Task 1 | Secure Architecture Design |
| Task 2 | Authentication & Authorization |
| Task 3 | Threat Modeling |
| Task 4 | Risk Mitigation |
| Task 5 | Resilience Evaluation |
| Cloud Platforms | Private |

# Cloud Computing

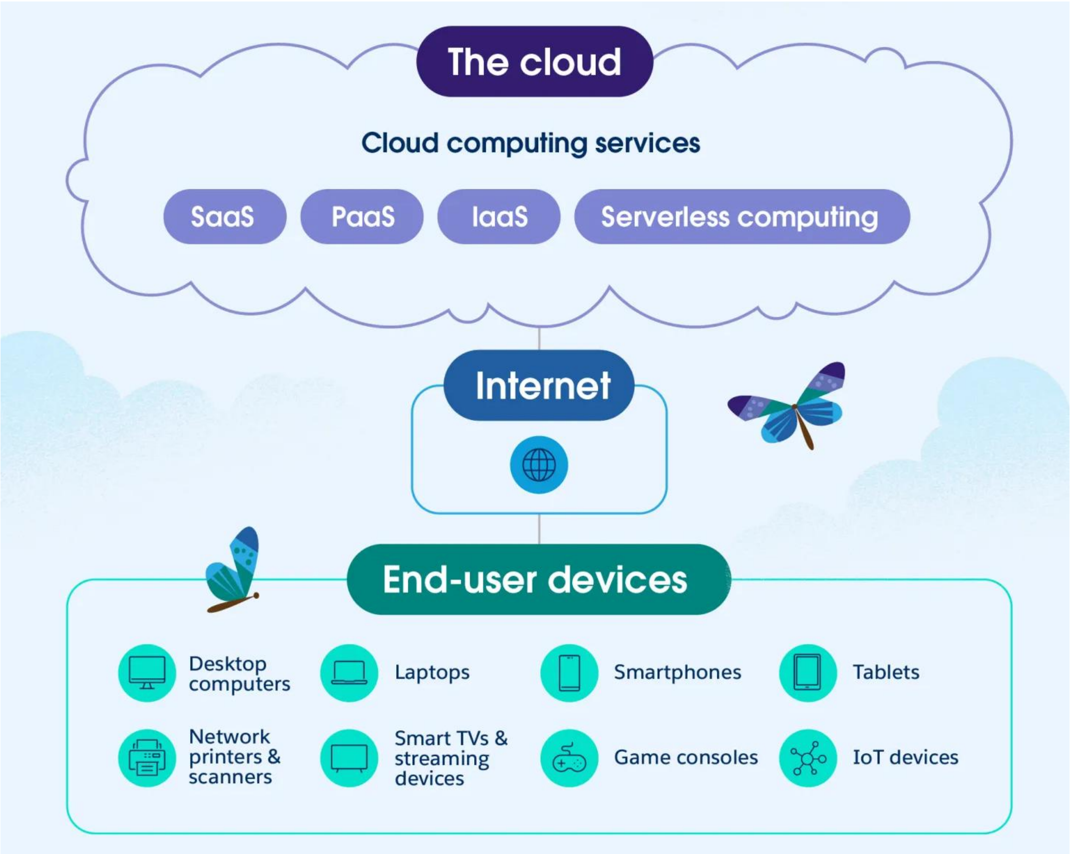
Cloud computing is the on-demand delivery of computing resources like servers, storage, databases, networking, and software over the internet

## Private cloud

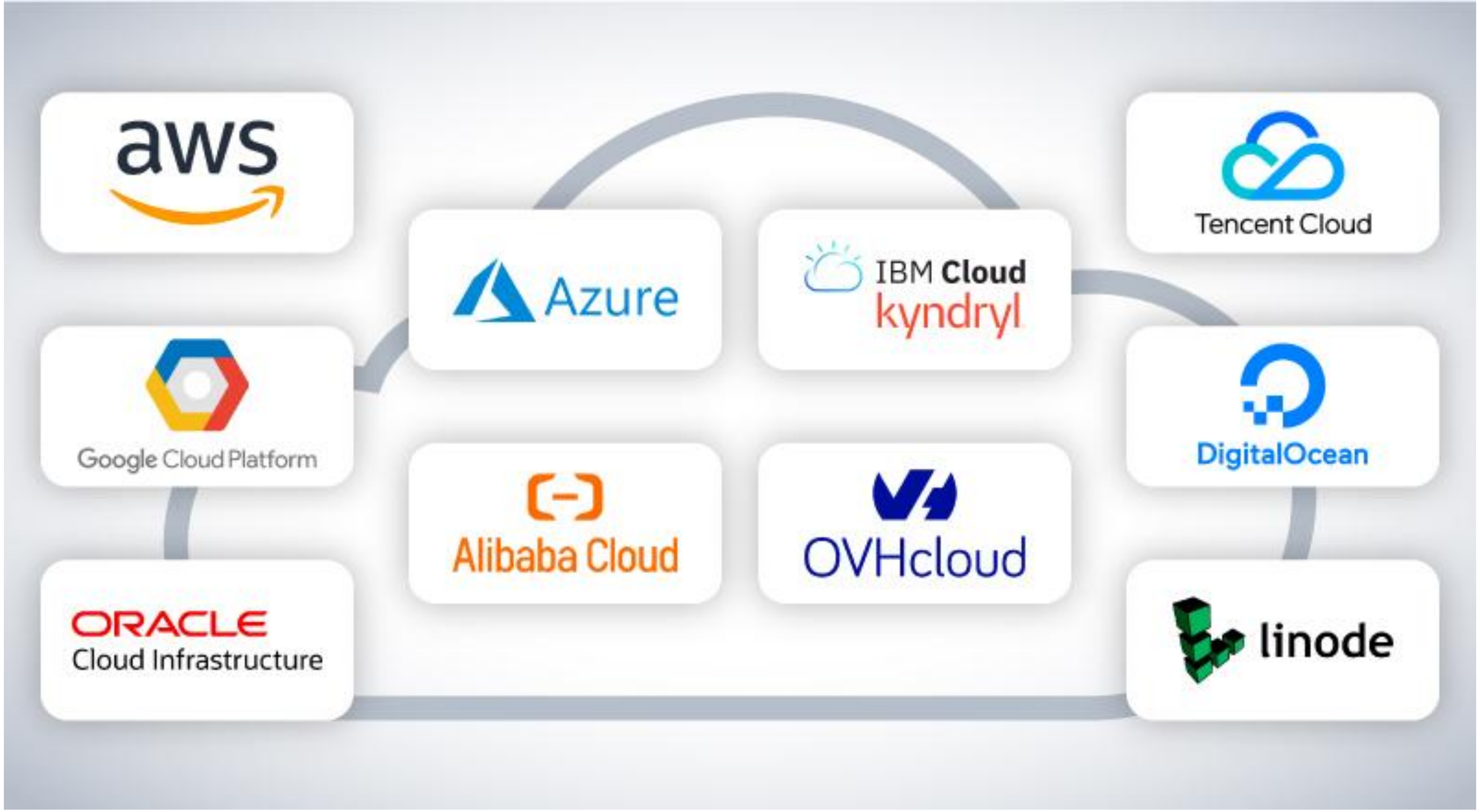
In this model, the service and infrastructure is dedicated solely to one organization

## Public cloud

In a public cloud, services and infrastructure are hosted by third-party cloud providers and shared among multiple customers.



# Cloud Platforms



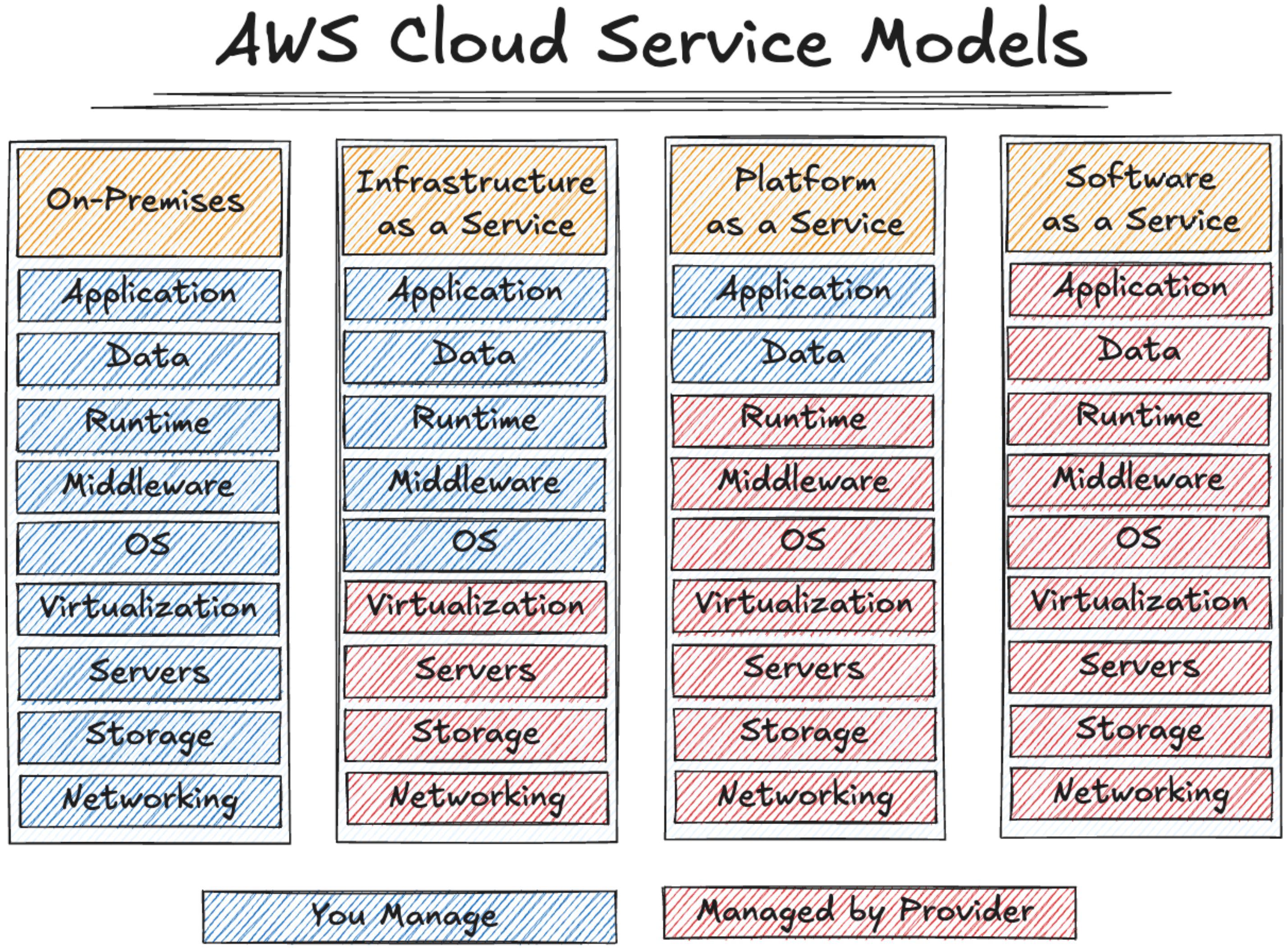
# Introduction to AWS Cloud Platform


- ✓ Create Free Login
- ✓ Try Creating Free Amazon EC2 virtual machine
- ✓ Access it from your Windows/ Linux

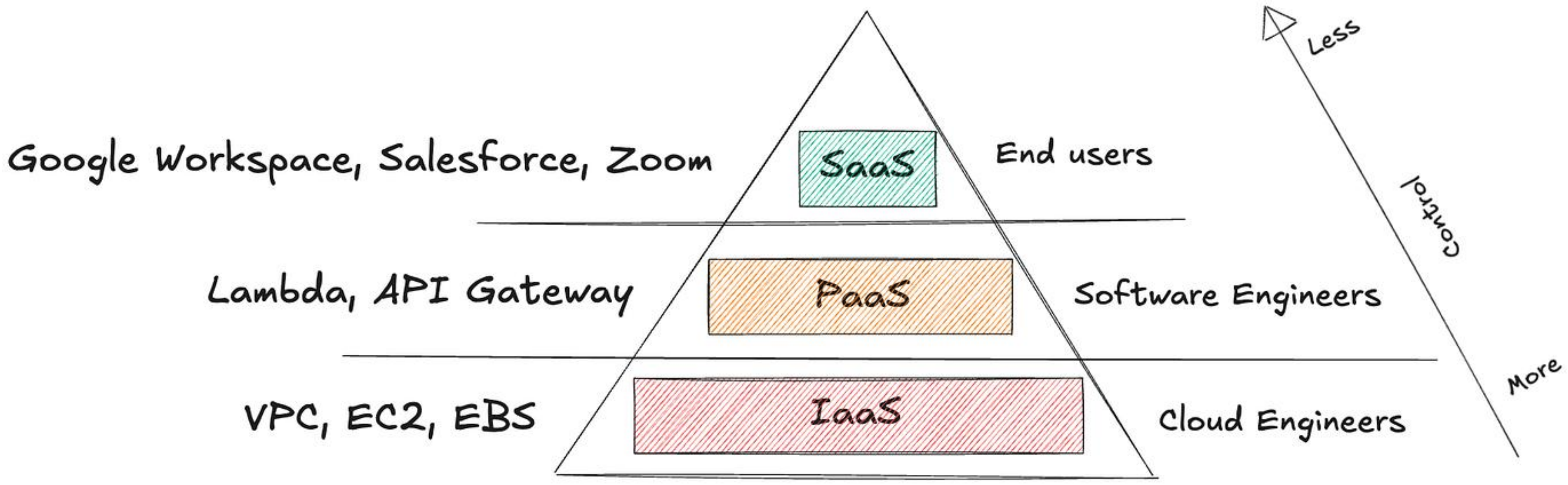
Demo

https://www.youtube.com/watch?v=6bb-Oqbl8-E

# Introduction to AWS Cloud Services



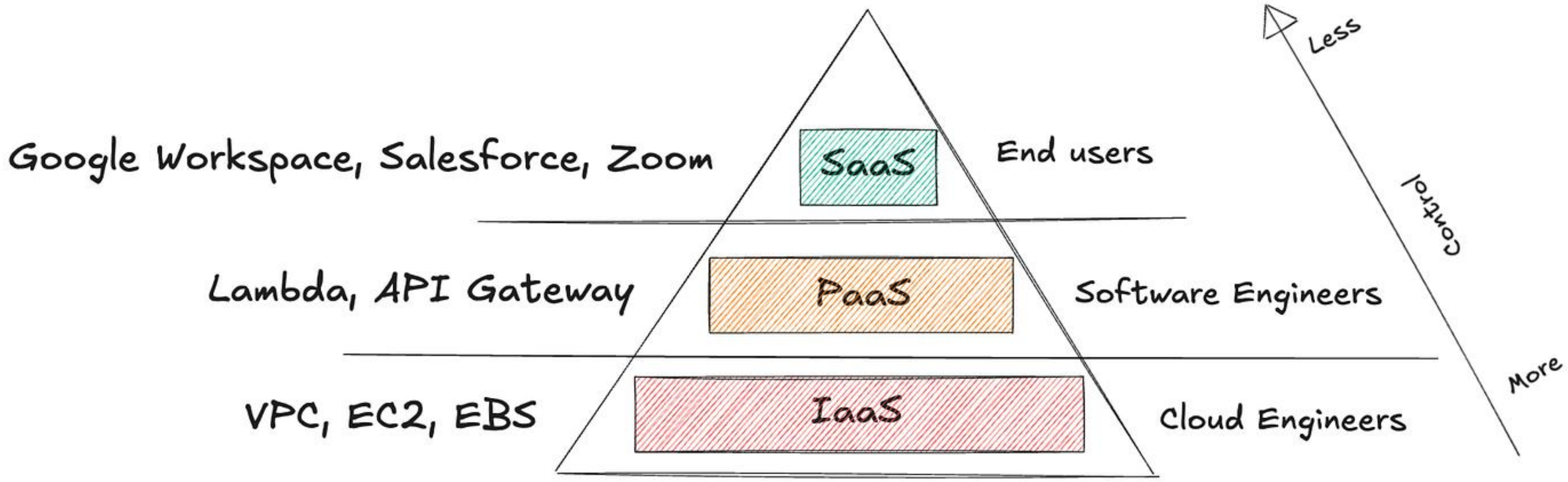
# AWS- IaaS




- ✓ **Virtual Private Cloud (VPC):** Isolated network for managing IP ranges, subnets, and access control.
- ✓ **Elastic Compute Cloud (EC2):** Virtual servers to run applications.
- ✓ **Elastic Block Storage (EBS):** Persistent storage for EC2 instances.

**Demo**

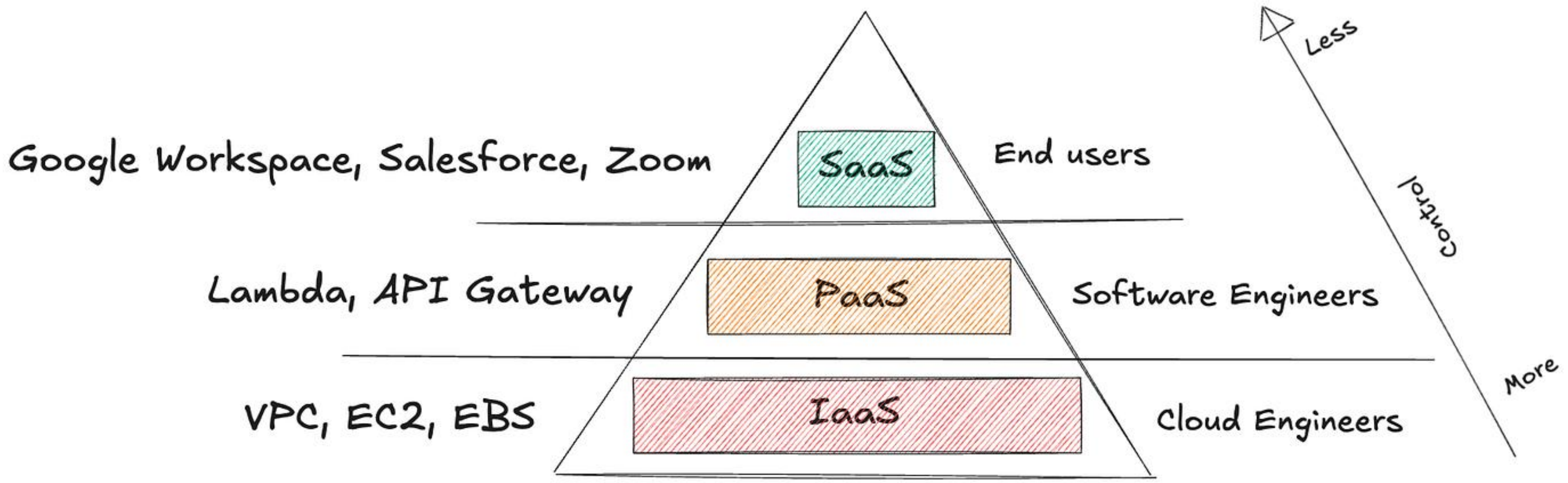
# AWS- PaaS




- ✓ **Elastic Beanstalk**: A fully managed service that makes it easy to deploy and scale web applications and services.
- ✓ **RDS (Relational Database Service)**: A managed service for setting up, operating, and scaling databases without having to worry about database maintenance, patching, or backups.
- ✓ **Lambda**: A serverless computing service that lets you run code in response to events without provisioning or managing servers.
- ✓ **API Gateway**: A fully managed service that makes it easy to create, publish, and manage APIs at scale.

**Demo**

# IaaS




•**Google Workspace:** A cloud-based suite of collaboration tools like Gmail, Calendar, Meet, Chat, Drive, Docs, Sheets, and Slides — a great choice for team collaboration.

•**Zoom:** A communication platform that can be used for video conference meetings, audio conferences, webinars, and live chat.

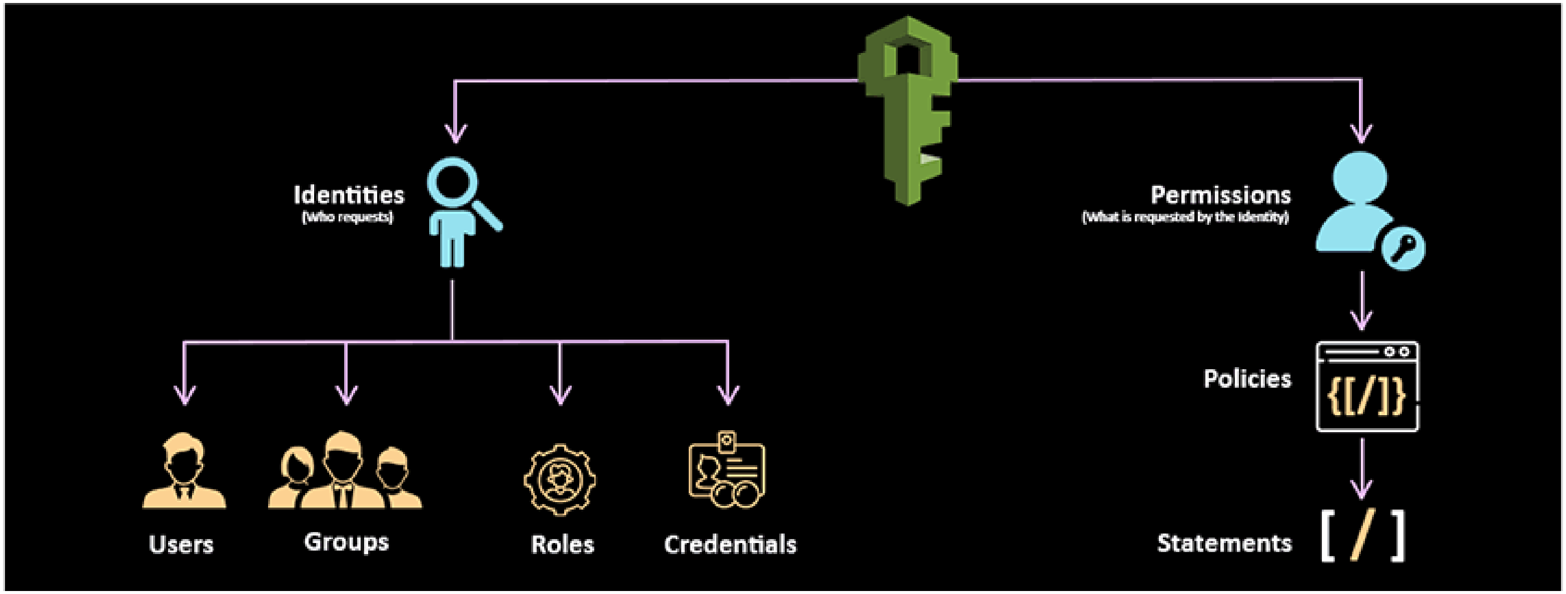
# Introduction to AWS Security Tools

# Introduction to IAM (Identity and Access Management)

- ✓ A core **secure access service** used in cloud platforms like AWS
- ✓ You should not use root user for using AWS account

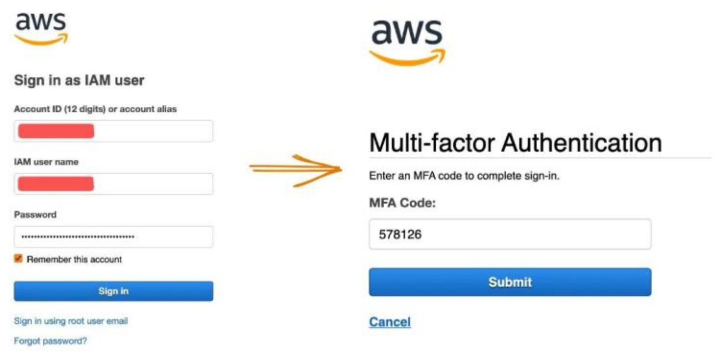
**ROOT USER** →

**Controlling permission of the AWS account**



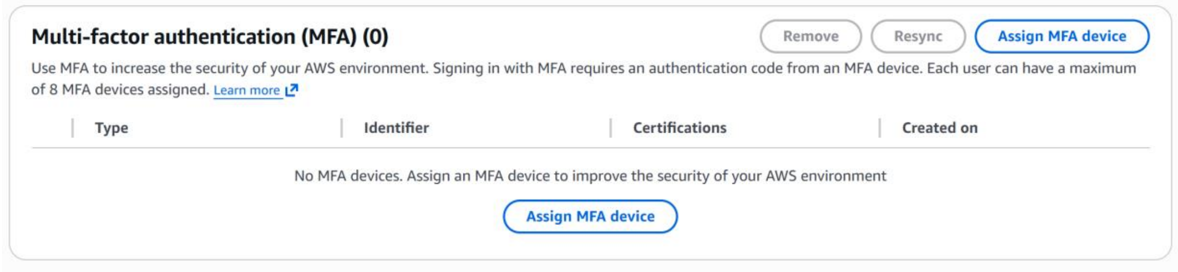
# Introduction to IAM (Identity and Access Management)

## MFA (Multi-Factor Authentication)




Login requires **more than just a password**.

Even if someone steals your password, they **still cannot log in**.

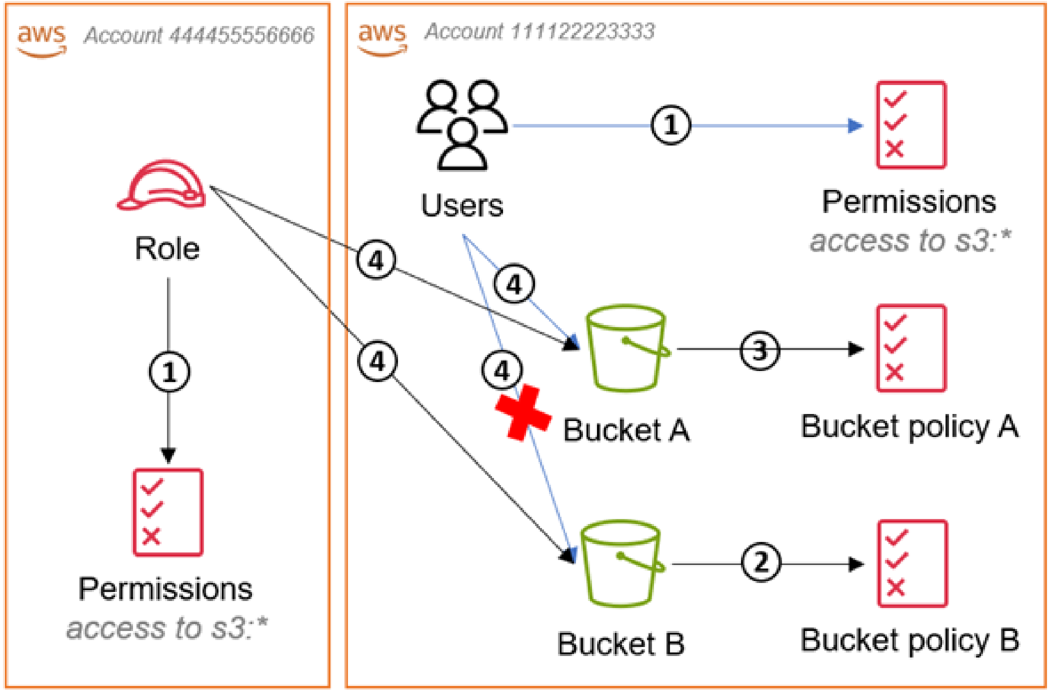



**JWT (JSON Web Token)** is a **secure token** given to a user after login.

# Introduction to IAM (Identity and Access Management)

## Fine-Grained Authorization and Access Control

Give *very specific permissions* to users - down to individual actions and resources.




## Manages Users and Their Access

Create users: Alice, Bob, Admin

Each user gets **unique credentials and permissions**.

## Manage Roles and Permissions

Role: EC2-Admin → can start/stop virtual machines

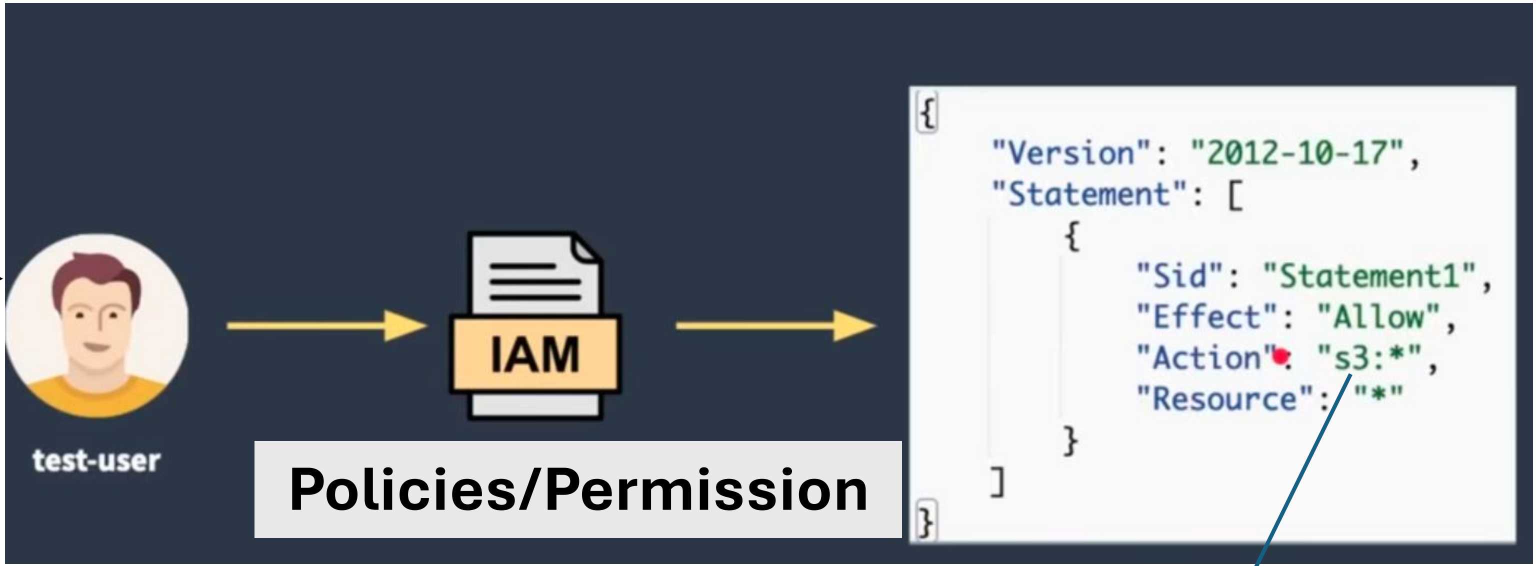
Role: S3-ReadOnly → can only view storage data

**Role-Based Access Control (RBAC)**

# USE CASE

**ROOT USER**

**Controlling permission of the AWS account**




**Storage/Bucket**

https://www.youtube.com/watch?v=bO25vbkoJlA

**Demo**

# *Thank You*

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